

React Front-End Curriculum

Part of the [training document series](#). This folder is a self-contained curriculum for building a decoupled, component-based frontend that consumes a backend API — the "Option B" presentation-layer approach described in the [architecture overview](#).

Reading order

1. **Core Concepts** — components, props, state, hooks, the custom-hook pattern for data fetching, and how a design system (tokens, themes, components, tests) is structured and consumed.
2. **Routing & API Integration** — client-side routing, a typed API client, mirroring backend response shapes, state-management options beyond local component state, and build-tooling concerns.
3. **Testing** — Vitest/Jest, React Testing Library, mocking, testing custom hooks, testing a data-heavy dashboard's patterns, and where end-to-end testing fits.
4. **Deployment** — static hosting, containerized deployment, CI/CD for a frontend build, service-worker caching and edge-gated routes, and the case where the build itself is the release.
5. **Patterns for Data-Heavy Dashboards** — view-model builders, independent sections, URL-backed state, a page-wide shared control, phone-sized variants, accessibility, theming, and embedding a conversational agent's answers.

Prerequisites

Comfort with modern JavaScript (functions, array methods, `async / await`, ES modules) is assumed. TypeScript examples are used throughout, but prior TypeScript experience is not required — the type annotations are explained as they're introduced.

See also

- [Architecture Overview](#) for how this compares to a single-process dashboard approach
- [Python API Backend](#) for the backend this frontend typically consumes
- [Data Science / ML](#) for what the underlying data being visualized typically represents
- [Conversational Agents](#) for surfacing an open-ended, natural-language view alongside the fixed charts/controls this curriculum builds